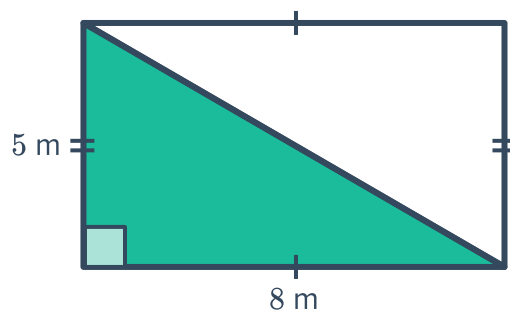


Triangles and rectangles



1 Consider the rectangle shown:

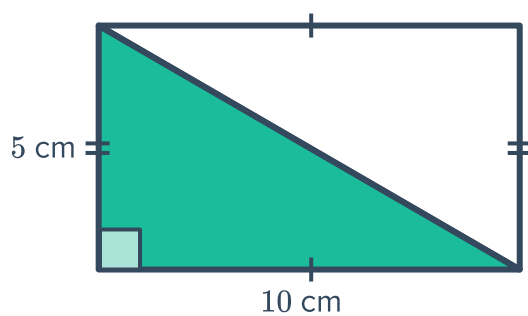
- a Find the area of the entire rectangle.
- b Find the area of the triangle.
- c What do you notice about the two areas?



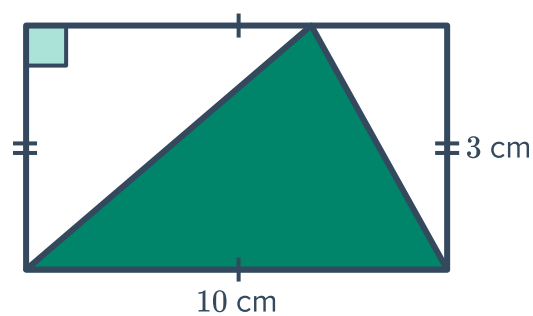
2 For each of the following figures:

- i Find the area of the entire rectangle.
- ii Find the area of the shaded triangle.

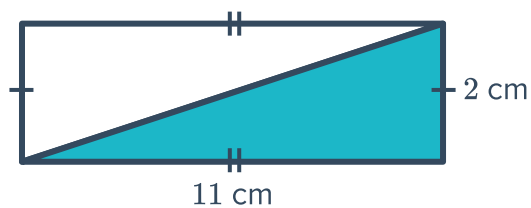
a



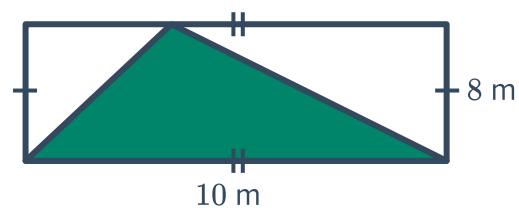
b



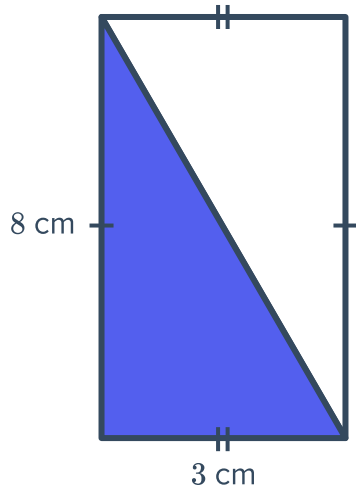
c



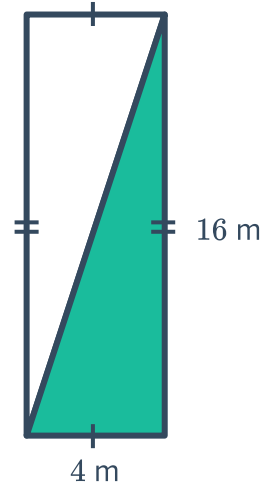
d



e

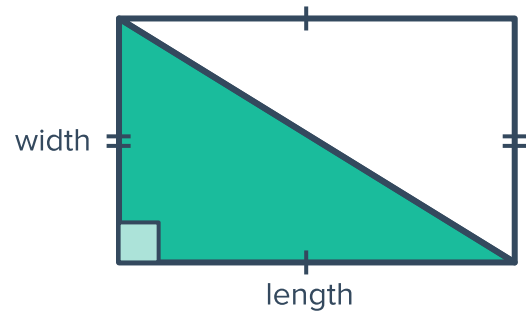


f



3 Consider the following triangle:

Which dimension of the following rectangle forms the height of the inside triangle?

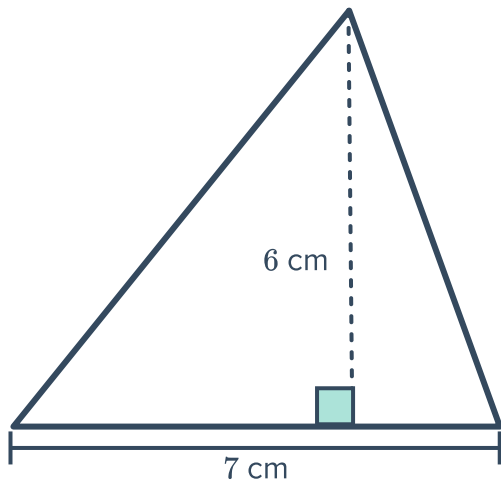




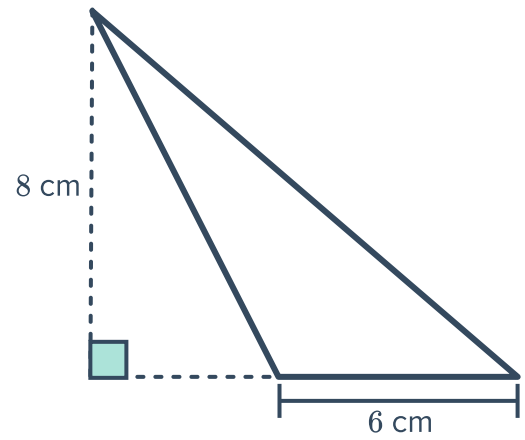
Area of a triangle

4 Find the area of the following triangles:

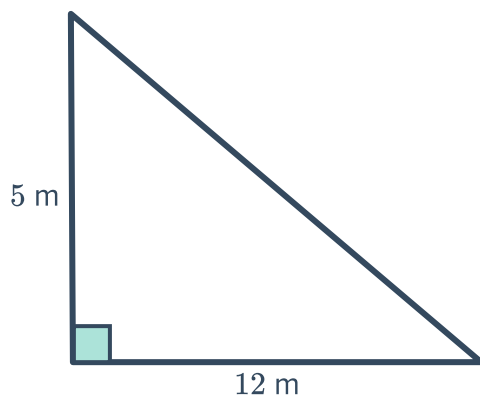
a



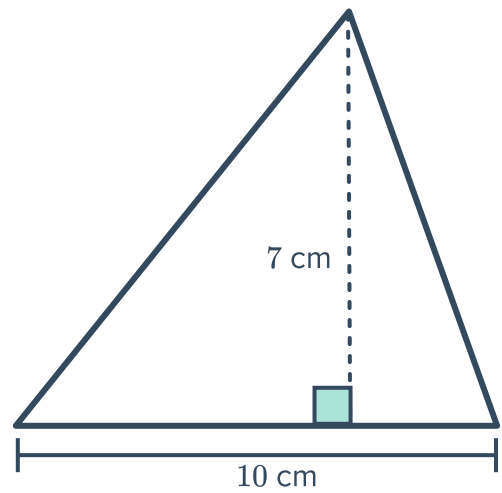
b



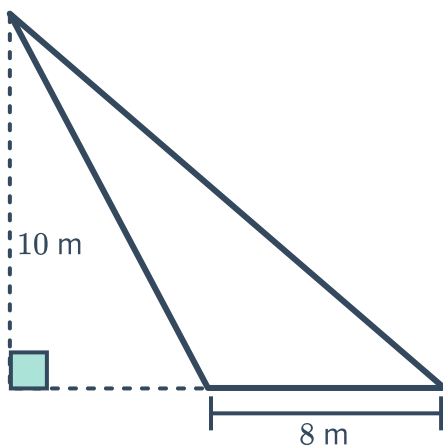
c



d



e



- 
- 5 Find the area of a triangle which has the following dimensions:
- a Height = 8 cm and base = 9 cm.

b Height = 4 m and base = 6 m.

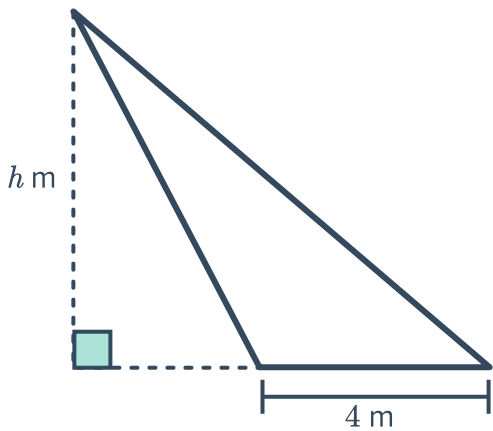
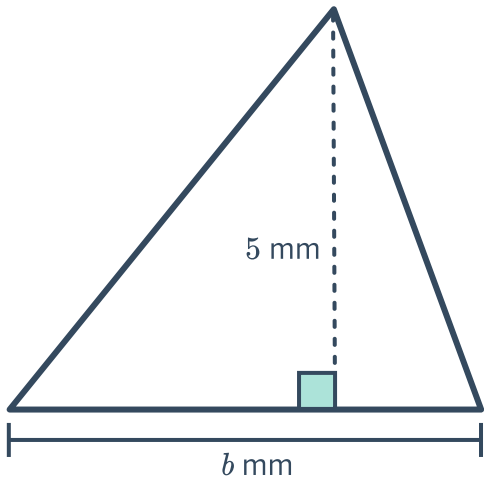
c Height = 7 mm and base = 8 mm.

d Height = 3 cm and base = 9 cm.

Dimensions from area

- 6 For each of the following triangles, find the value of b or h given the area:
- a Area = 20 mm^2

b Area = 48 m^2



- 7 Determine whether the following could be the dimensions of a triangle with an area of 20 mm^2 :
- a Base = 8 mm, height = 5 mm.

b Base = 1 mm, height = 20 mm.

c Base = 5 mm, height = 8 mm.

d Base = 2 mm, height = 20 mm.
- 8 Complete the following table of base and height measurements for different triangles, which all have an area of 30 m^2 :

Base (m)		10	15
Height (m)	12		
Area (m^2)	30	30	30

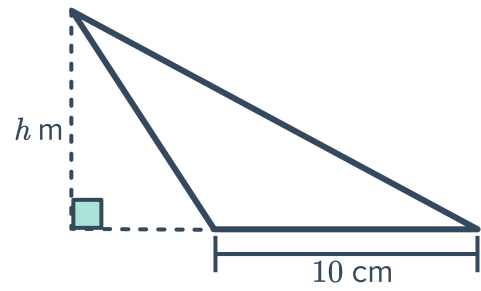
Applications

- 9 Lisa has purchased a rectangular piece of fabric measuring 6 m in length and 9 m in width.



- 10 A gutter running along the roof of a house has a cross-section in the shape of a triangle as shown:

If the area of the cross-section is 40 cm^2 , and the length of the base of the gutter is 10 cm , find the perpendicular height h of the gutter.



- 11 The faces on a 4-sided die are all triangular. Each face has a base length of 19 mm and a perpendicular height of 12 mm . Find the area of one face.